

Tautology :-

A compound proposition that is always true is called a tautology.

e.g.:- $P \vee \neg P$

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

Contradiction :-

A compound proposition that is always false is called a contradiction.

e.g.:- $P \wedge \neg P$

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

Contingency :-

A compound proposition that is neither a tautology nor a contradiction is called a contingency.

e.g.:- $P \rightarrow Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

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Ques:- Check the following compound proposition are tautology or contradiction or contingency:

(1) $[P \wedge (P \rightarrow Q)] \rightarrow Q$

(2) $[(P \rightarrow Q) \wedge (Q \rightarrow R)] \rightarrow (P \rightarrow R)$

(3) $Q \vee (\neg Q \wedge P)$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg(P \vee Q)$	$P \wedge \neg Q$	$\neg P \wedge Q$
T	T	F	F	T	F	F	F
T	F	F	T	T	F	T	F
F	T	T	F	T	F	F	T
F	F	T	T	F	T	F	F

Logical Equivalences

Compound propositions that have the same truth values in all possible cases are called logically equivalent.

OR

The compound proposition p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology.

The notation $p \equiv q$ denotes that p and q are logically equivalent. ($\equiv$ or $\Leftrightarrow$)

Ques :- Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$	$\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$
T	T	F	F	T	F	F	T
T	F	F	T	T	F	F	T
F	T	T	F	T	F	F	T
F	F	T	T	F	T	T	T

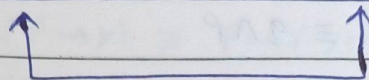
Since, truth values of the compound propositions $\neg(p \vee q)$ and $\neg p \wedge \neg q$ agree for all possible combinations of the truth values of p and q , it follows that $\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$ is a tautology and so given compound propositions are equivalent.

De Morgan's Laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

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Eg* Show that $P \rightarrow Q \equiv \neg P \vee Q$.

P	Q	$P \rightarrow Q$	$\neg P$	$\neg P \vee Q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T



Note :-

In general, 2^n rows are required if a compound proposition involves n propositional variables.

Logical Equivalences

Identity Laws :-

$$P \wedge T \equiv P$$

$$P \vee F \equiv P$$

Domination Laws :-

$$P \vee T \equiv T$$

$$P \wedge F \equiv F$$

Idempotent Laws :-

$$P \vee P \equiv P$$

$$P \wedge P \equiv P$$

Double negation Law :-

$$\neg(\neg P) \equiv P$$

Commutative Laws :-

$$P \vee Q \equiv Q \vee P$$

$$P \wedge Q \equiv Q \wedge P$$

Associative Laws :-

$$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$$

$$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$$

Distributive Laws :-

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

De Morgan's Laws :-

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Absorption Laws :-

$$P \vee (P \wedge Q) \equiv P$$

$$P \wedge (P \vee Q) \equiv P$$

Negation Laws :-

$$P \vee \neg P \equiv T$$

$$P \wedge \neg P \equiv F$$

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Logical equivalences involving conditional statements

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$$

$$P \vee Q \equiv \neg P \rightarrow Q$$

$$P \wedge Q \equiv \neg(P \rightarrow \neg Q)$$

$$\neg(P \rightarrow Q) \equiv P \wedge \neg Q$$

$$(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R)$$

$$(P \rightarrow R) \wedge (Q \rightarrow R) \equiv (P \vee Q) \rightarrow R$$

$$(P \rightarrow Q) \vee (P \rightarrow R) \equiv P \rightarrow (Q \vee R)$$

$$(P \rightarrow R) \vee (Q \rightarrow R) \equiv (P \wedge Q) \rightarrow R$$

Ques :- Show that $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$.

Solⁿ :- $\neg(P \rightarrow Q) \equiv \neg(\neg P \vee Q)$

$$\equiv \neg(\neg P) \wedge \neg Q \quad (\text{by De Morgan Law})$$

$$\equiv P \wedge \neg Q \quad (\text{by double negation Law})$$

Ques :- Show that $\neg(P \vee (\neg P \wedge Q))$ and $\neg P \wedge \neg Q$ are logically equivalent by developing a series of logical equivalences.

Solⁿ :- $\neg(P \vee (\neg P \wedge Q)) \equiv \neg P \wedge \neg(\neg P \wedge Q)$ (by De Morgan Law)

$$\equiv \neg P \wedge [\neg(\neg P) \vee \neg Q] \quad (\text{by De Morgan Law})$$

$$\equiv \neg P \wedge (P \vee \neg Q) \quad (\text{by double negation Law})$$

$$\equiv (\neg P \wedge P) \vee (\neg P \wedge \neg Q) \quad (\text{by distributive Law})$$

$$\equiv F \vee (\neg P \wedge \neg Q) \quad (\because \neg P \wedge P = F) \quad \text{Negation Law}$$

$$\equiv \neg P \wedge \neg Q \quad (\text{by identity Law})$$

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Ques:- Show that $(P \wedge Q) \rightarrow (P \vee Q)$ is a tautology.

Sol.ⁿ :- $(P \wedge Q) \rightarrow (P \vee Q) \equiv \neg(P \wedge Q) \vee (P \vee Q)$
 $\equiv (\neg P \vee \neg Q) \vee (P \vee Q)$ (by De Morgan Law)
 $\equiv (\neg P \vee P) \vee (\neg Q \vee Q)$ (by Associative and comm. Law)

$$\equiv T \vee T$$

$$\equiv T$$

Well formed formula :-

A grammatically correct expression is called a well formed formula. Every atomic statement (proposition) is a well formed formula.

* If P is WFF then $\neg P$ is also WFF.

* If P and Q are WFF then $P \wedge Q$, $P \vee Q$, $P \rightarrow Q$ are also WFF.

Tautological implication :-

A compound proposition A is said to be tautologically imply the compound proposition B iff $A \rightarrow B$ is a tautology.

e.g :- (1) $P \wedge Q$ tautologically implies Q .

i.e. $(P \wedge Q) \rightarrow Q$ will be a tautology.

(2) $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow R)$ tautologically implies R